

Task 6

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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
df = pd.read_csv("C:/Users/JAAVANIKA L/fall semester 22-23/Downloads/Iris.csv")
```

```
df.head()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
X = df.iloc[:, 0:2].values
y = df['Species'].astype('category').cat.codes
```

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y = df['Species'].astype('category').cat.codes
```

```
scaler = StandardScaler()
X = scaler.fit_transform(X)
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=0)
```

```
model = KNeighborsClassifier(n_neighbors=3)
model.fit(X_train, y_train)
```

```
▼ KNeighborsClassifier ⓘ ?
KNeighborsClassifier(n_neighbors=3)
```

```
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

Accuracy: 0.9777777777777777

Confusion Matrix:

```
[[16  0  0]
 [ 1 17  0]
 [ 0  0 11]]
```

```
def plot_decision_boundary(model, X, y):
    h = 0.02
    x_min, x_max = X[:, 0].min() - 1, X[:, 0].max() + 1
    y_min, y_max = X[:, 1].min() - 1, X[:, 1].max() + 1
    xx, yy = np.meshgrid(np.arange(x_min, x_max, h),
                          np.arange(y_min, y_max, h))
    Z = model.predict(np.c_[xx.ravel(), yy.ravel()])
    Z = Z.reshape(xx.shape)
    plt.contourf(xx, yy, Z, alpha=0.3)
    plt.scatter(X[:, 0], X[:, 1], c=y, edgecolors='k')
    plt.title("Decision Boundary (K=3)")
    plt.xlabel("SepalLengthCm")
    plt.ylabel("SepalWidthCm")
    plt.show()
```

```
plot_decision_boundary(model, X, y)
```

